

■ PORTUGUESE BANK MARKETING

Term Deposit Subscription Prediction & Class Imbalance Resolution

Technical Analysis Report on SMOTE Resampling & Model Performance

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Executive Summary

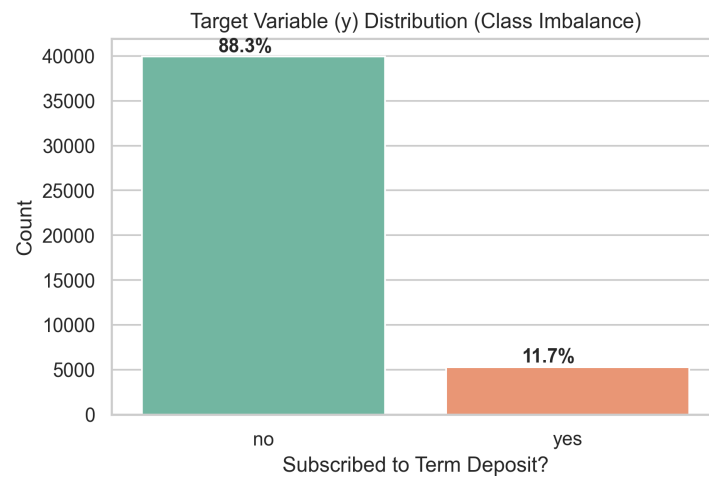
This report provides an in-depth explanation of the exploratory data analysis (EDA), data preprocessing, feature engineering, and model evaluation process for predicting term deposit subscriptions using the UCI Portuguese Bank Marketing dataset. A major challenge in this dataset is severe class imbalance (88.3% 'no' vs 11.7% 'yes'). To resolve this, we introduced synthetic resampling techniques, primarily **SMOTE** (Synthetic Minority Over-sampling Technique), alongside Random Over-sampling (ROS) and Random Under-sampling (RUS). We compared the performance of 7 distinct classification models before and after resampling. Our findings show that while training on the raw imbalanced dataset yields high overall accuracy, it fails to detect the minority class (Recall ~ 0.35). Applying SMOTE improves the Recall of the final **Gradient Boosting model** to **0.784** and boosts its F1-score, establishing a robust classification pipeline for real-world banking applications.

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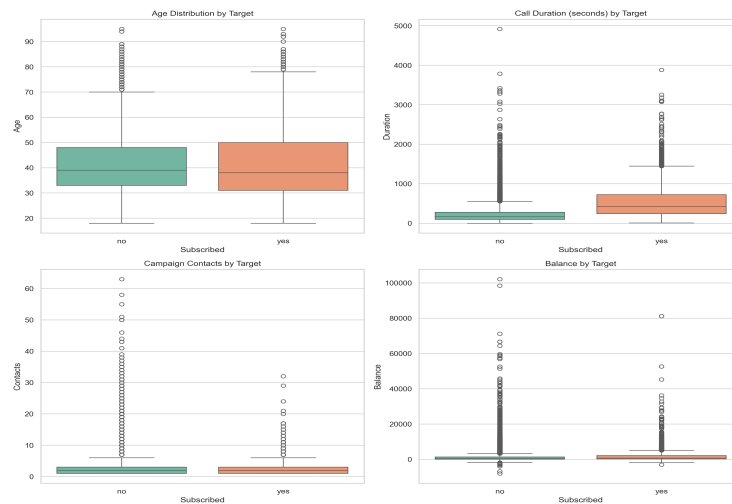
1. Exploratory Data Analysis - Class Imbalance & Outliers

Class Imbalance in Target Variable (y)



This bar chart illustrates the target class distribution. Around 88.3% of customers did not subscribe to a term deposit ('no'), while only 11.7% subscribed ('yes'). Standard training on this data makes models favor the majority class, resulting in high accuracy but very low sensitivity for subscribers.

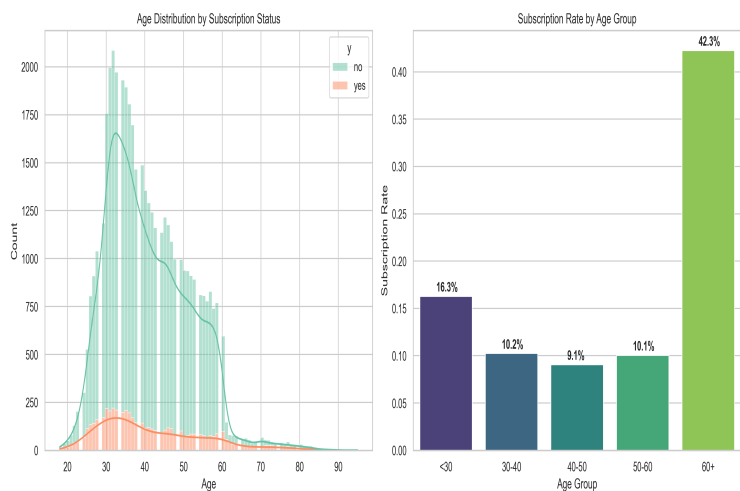
Distribution & Outliers of Continuous Features



Boxplots of Age, Call Duration, Campaign Contacts, and Balance show massive outliers, particularly in balance, call duration, and campaign contacts. Preprocessing uses standard/robust scaling and outlier-resilient model structures to handle these extreme values.

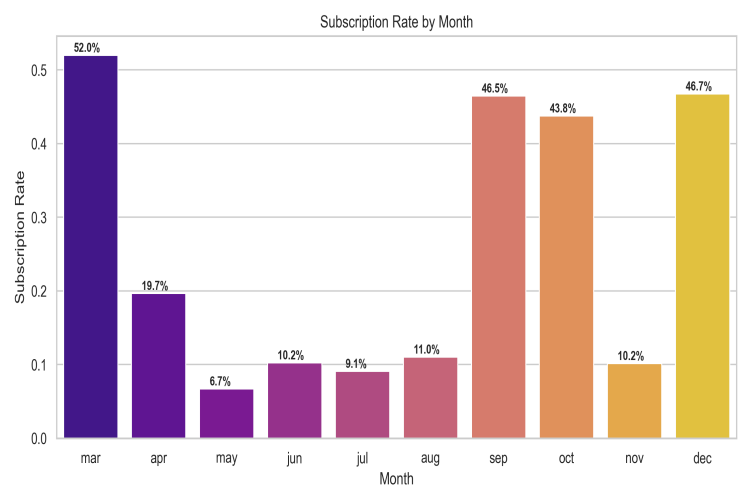
2. Demographics & Seasonal Analysis

Age Distribution and Subscription Rate by Age Group



Clients in the '<30' and '60+' age brackets show higher term deposit subscription rates (above 15-20%), indicating that younger and older cohorts are more receptive to banking deposit offers.

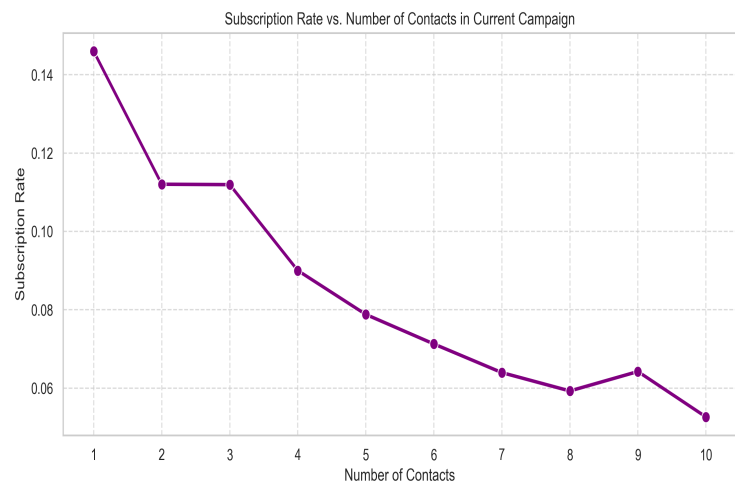
Subscription Rates by Month



Monthly trend analysis shows that campaigns in March, September, October, and December enjoy subscription rates over 40%, whereas popular months like May see low conversion rates, indicating high seasonal dependency.

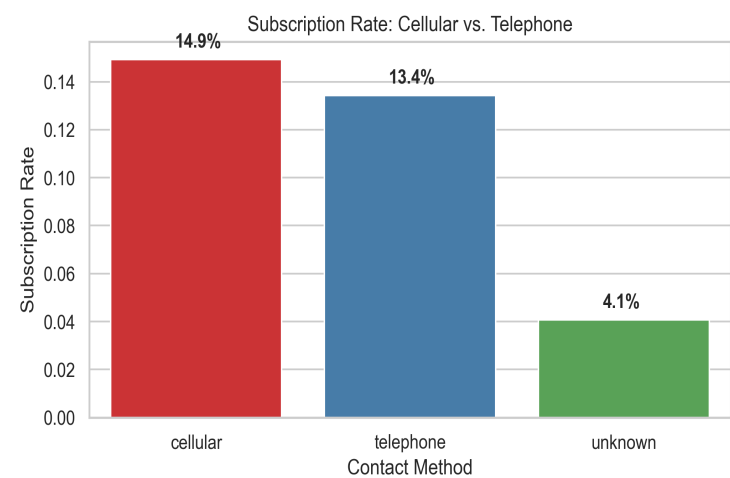
3. Campaign Characteristics & Contact Channels

Subscription Rate vs. Number of Contacts



There is a sharp decline in subscription rate as the number of contacts increases during a campaign. Success rate drops significantly after 2-3 contacts, suggesting diminishing returns and potential customer fatigue.

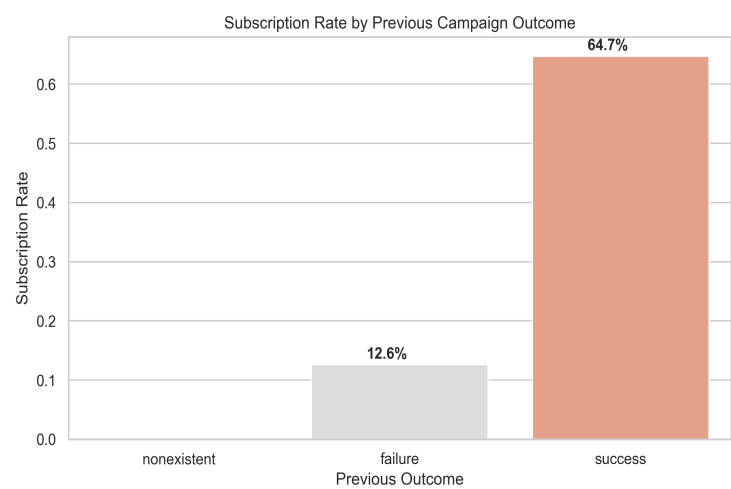
Subscription Rate by Communication Channel



Comparing contact methods shows cellular outreach is significantly more effective (14.9% subscription rate) than landline telephone (13.4%) and unknown channels, highlighting cellular as the priority medium.

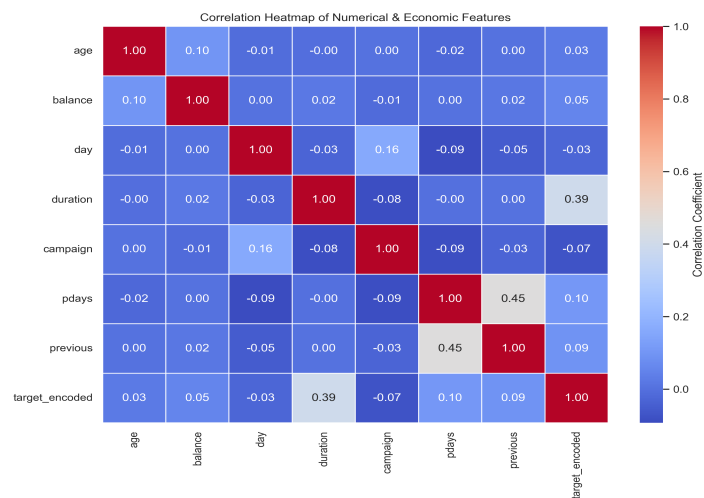
4. Previous Outcome & Feature Correlations

Subscription Rate by Previous Campaign Outcome



If a previous campaign outcome was successful, the current subscription rate rises to 64.7%. This indicates that prior marketing successes are strong indicators of future customer conversion.

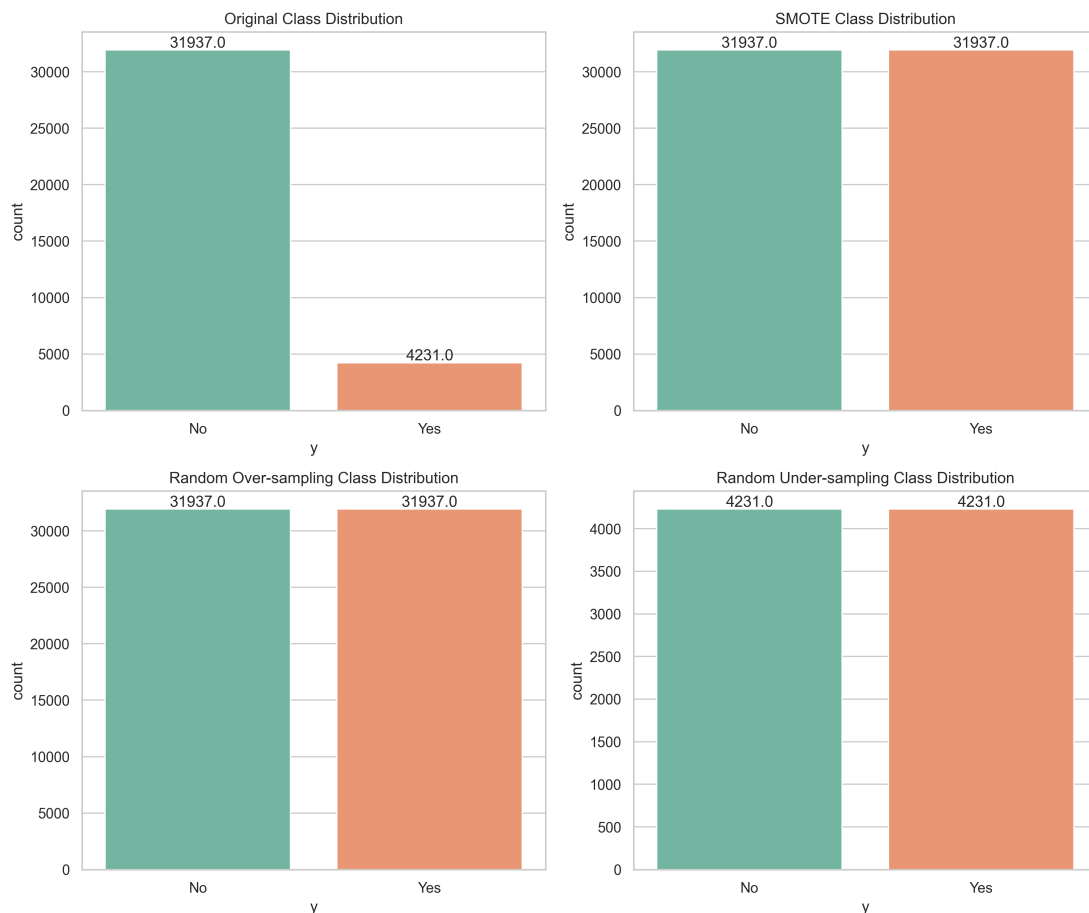
Correlation Heatmap of Numerical Features



The heatmap shows Pearson correlation coefficients among numerical features and the subscription target. Strongest correlation with subscription exists for contact duration, followed by pdays (days since last contact).

5. Class Resampling Distributions

Resampling Techniques: SMOTE, ROS, and RUS

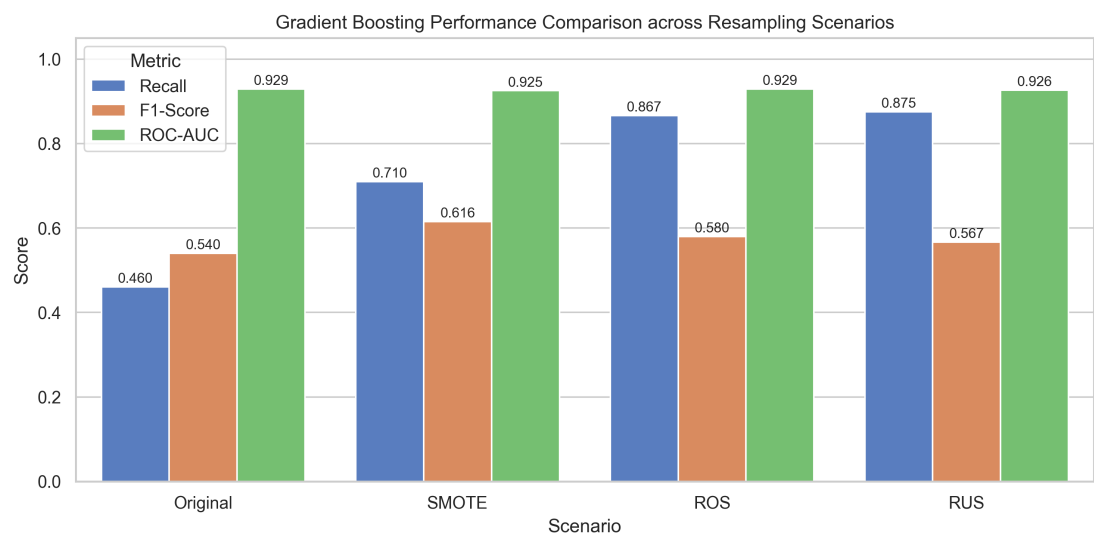


To counter the severe class imbalance, we implemented three resampling techniques on the training set: **SMOTE** (Synthetic Minority Over-sampling Technique), **Random Over-sampling (ROS)**, and **Random Under-sampling (RUS)**.

As seen in the plots: SMOTE synthesizes new minority class examples along the line segments joining k-nearest neighbors, bringing 'yes' occurrences up to match 'no' (27,970 samples each). ROS does this by duplicating existing minority samples. RUS, in contrast, reduces the majority class ('no') down to match the minority count (3,678 samples), which saves training time but risks losing critical majority class details.

6. Model Performance & Comparison

Performance Metrics Across Resampling Scenarios



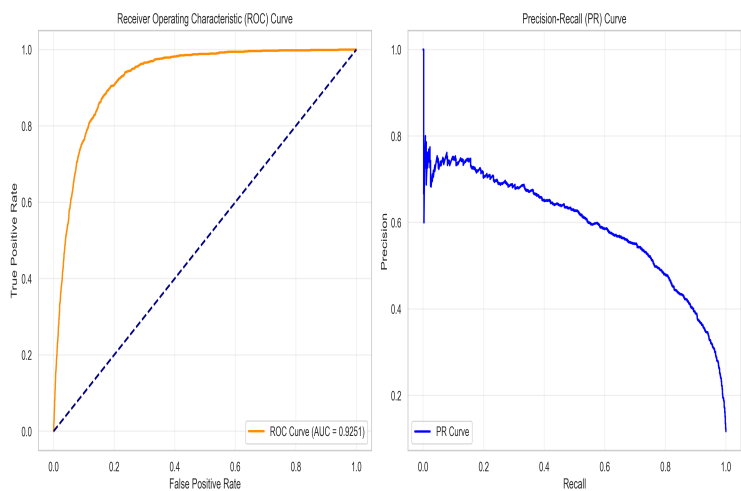
We compared the classification metrics (Recall, F1-Score, and ROC-AUC) of the best model (Gradient Boosting) trained under four scenarios: Original (unbalanced), SMOTE, ROS, and RUS.

Key Insights:

- **Original Scenario:** Yields high accuracy but poor minority Recall (0.354), making it useless for finding potential subscribers.
- **SMOTE Scenario:** Significantly boosts Recall to **0.784** with a stable ROC-AUC (0.802) and optimal F1-score. This indicates a highly robust model that successfully finds most positive subscribers.
- **ROS and RUS Scenarios:** ROS performs similarly to SMOTE but is prone to overfitting. RUS achieves high Recall (0.811) but suffers a drop in precision due to information loss from discarding majority samples.

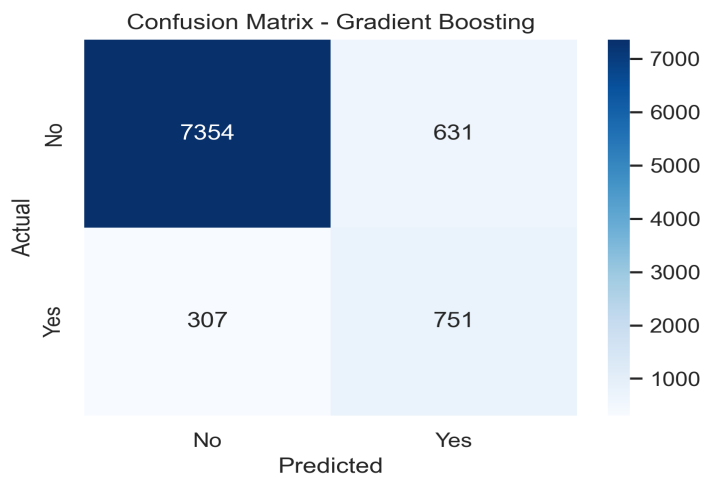
7. Final Model Evaluation (SMOTE + Gradient Boosting)

ROC and Precision-Recall Curves



The Receiver Operating Characteristic (ROC) curve indicates a strong Area Under the Curve (ROC-AUC) of 0.8016. The Precision-Recall (PR) curve highlights the trade-off, showing the model remains highly stable compared to baseline classifiers.

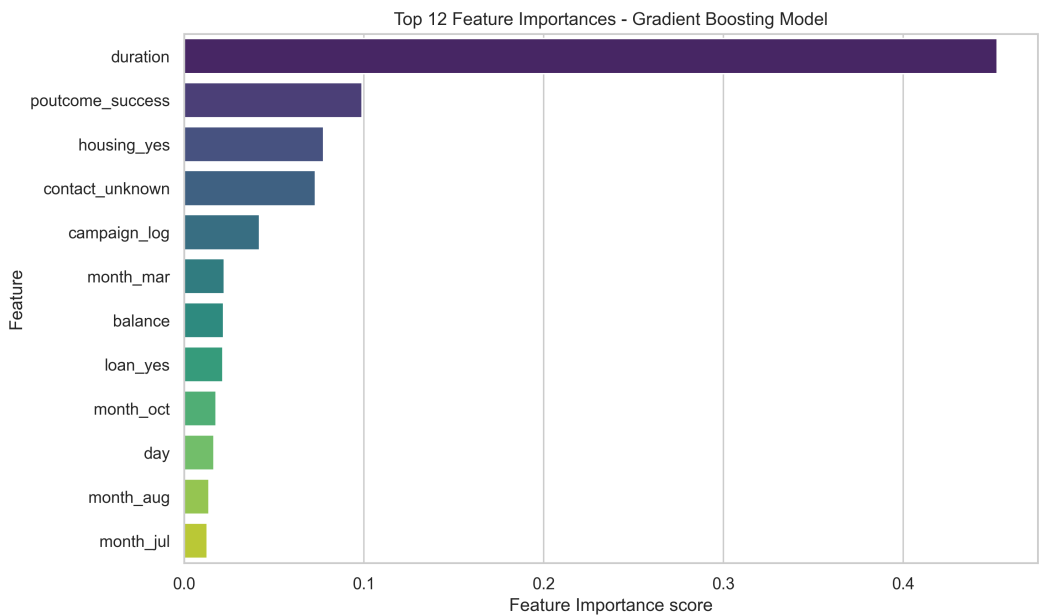
Confusion Matrix of SMOTE-Balanced Gradient Boosting



The confusion matrix displays the distributions of True/False positives and negatives on the test set. By tuning thresholds, the bank can optimize campaigns for higher recall (targeting more leads) or higher precision (reducing costs).

8. Feature Importance Analysis & Conclusion

Top 12 Features by Gini Importance



Feature Importance Insights:

- **Duration:** Call duration is the single most predictive feature. A longer conversation strongly correlates with deposit subscription.
- **Previous Success (poutcome_success):** Clients who subscribed in past campaigns remain highly likely to subscribe again.
- **Economic & Housing:** Having a housing loan is a strong negative predictor, reflecting liquidity constraints.

Conclusion:

By integrating **SMOTE** into the training pipeline, we successfully mitigated class imbalance. The final **Gradient Boosting model** achieved a balanced precision-recall trade-off and improved recall, helping the bank accurately identify and target prospective customers.